Chapter 5. Partitions

Divide and Conquer

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Chapter 3 counted lists. Here we count *distributions* of n balls into k boxes. Balls may be identical or distinct, and so may boxes, which gives compositions, set partitions and integer partitions.

5.1 Compositions

Motivation. Identical balls into distinct boxes: only the number of balls per box matters, so we count ways to write n as an ordered sum. Lining up balls and separating walls turns this into a multiset permutation.

Definition 5.1 (Compositions)

A sequence $(a_1, \dots, a_k)$ of integers with $a_i \geq 0$ and $a_1 + \cdots + a_k = n$ is a **weak composition** of n into k parts; if all $a_i > 0$, it is a **composition**.

Theorem 5.2

For positive integers n, k, the number of weak compositions of n into k parts is $\binom{n+k-1}{k-1} = \binom{n+k-1}{n}$.

Proof. This counts the ways to put n identical balls into k distinct boxes. Line up the boxes and record the contents as a row of n balls and $k-1$ walls. The row determines the distribution and vice versa, so we count permutations of a multiset with n balls and $k-1$ walls: $\frac{(n+k-1)!}{n!(k-1)!}$ by Theorem 3.5. ■



$3 + 0 + 4 + 1$

Figure. *Balls and walls: the weak composition* $(3, 0, 4, 1)$ *of* 8 *into* 4 *parts as an arrangement of* 8 *balls and* 3 *walls.*

Corollary 5.3

The number of compositions of n into k parts is $\binom{n-1}{k-1}$.

Proof [Supplement]. Subtracting 1 from every part is a bijection from compositions of n into k parts onto weak compositions of $n-k$ into k parts (give each box one ball first). By Theorem 5.2 there are $\binom{n-k+k-1}{k-1} = \binom{n-1}{k-1}$ of them. ■

Corollary 5.4

A positive integer n has 2^{n-1} compositions (into any number of parts).

Proof.

Step 1 (summing). A composition has between 1 and n parts, so by Corollary 5.3 and Theorem 4.4 the number is $\sum_{k=1}^{n} \binom{n-1}{k-1} = 2^{n-1}$.

Step 2 (doubling, alternative). By induction: 1 has one composition. From each composition C of n form two compositions of $n+1$: add 1 to the first part, or write a new part 1 in front. Every composition of $n+1$ arises exactly once (according to whether its first part is ≥ 2 or $= 1$), so $n+1$ has twice as many compositions as n. ■

(Weak compositions into an unspecified number of parts are infinite in number, since zeros can be appended.)

5.2 Set partitions

Motivation. Distinct balls into identical boxes means splitting $[n]$ into blocks. There is no closed formula, but the recurrences come from asking where the largest element goes, and the numbers relate powers to falling factorials.

Definition 5.5 (Set partitions, Stirling numbers of the second kind)

A **partition** of $[n]$ is a collection of nonempty **blocks** such that each element of $[n]$ lies in exactly one block. $S(n,k)$, the **Stirling number of the second kind**, is the number of partitions of $[n]$ into k blocks. $S(n,k) = 0$ for $n < k$; by convention $S(0,0) = 1$ and $S(n,0) = 0$ for $n > 0$.

Example 5.6

$S(n,1) = S(n,n) = 1$ for $n \geq 1$, and $S(n,n-1) = \binom{n}{2}$ for $n \geq 2$: such a partition consists of one doubleton and singletons.

Example 5.7

$S(4,2) = 7$: $\{1,2,3\}\{4\}$, $\{1,2,4\}\{3\}$, $\{1,3,4\}\{2\}$, $\{2,3,4\}\{1\}$, $\{1,2\}\{3,4\}$, $\{1,3\}\{2,4\}$, $\{1,4\}\{2,3\}$.

There is no closed formula for $S(n,k)$; a formula with one summation follows from the sieve (Chapter 7).

Theorem 5.8

For positive integers $k \leq n$:

$$S(n,k) = S(n-1,k-1) + k\,S(n-1,k). \tag{5.1}$$

Proof. Look at the element n. If it forms a singleton block, the others form a partition of $[n-1]$ into $k-1$ blocks: $S(n-1,k-1)$ ways. Otherwise, removing n leaves a partition of $[n-1]$ into k blocks, and n can be put into any of these k blocks: $kS(n-1,k)$ ways. ∎

Distinct balls into k distinct boxes, no box empty: partition $[n]$ into k blocks and label the blocks, giving $k!\,S(n,k)$.

Corollary 5.9

The number of surjections $f : [n] \to [k]$ is $k!\,S(n,k)$.

Proof. The fibres $f^{-1}(i)$, $i \in [k]$, form a partition of $[n]$ into k nonempty blocks labelled by $[k]$. Conversely, a partition into k blocks with a bijective labelling by $[k]$ defines a unique surjection. So the count is $S(n,k)$ times the $k!$ labellings. ∎

Corollary 5.10

For every real x and nonnegative integer n:

$$x^n = \sum_{k=0}^{n} S(n,k)(x)_k, \tag{5.2}$$

where $(x)_k = x(x-1)\cdots(x-k+1)$.

Proof. For $n = 0$ both sides are 1. For $n > 0$, $S(n,0) = 0$. Both sides are polynomials in x of degree n, so it suffices that they agree for infinitely many x, e.g. all positive integers. For $x \in \mathbb{N}$, x^n counts all functions $[n] \to [x]$ (Theorem 3.6). Group them by image: an image of size k can be chosen in $\binom{x}{k}$ ways and the function onto it in $k!\,S(n,k)$ ways (Corollary 5.9), and $\binom{x}{k}k! = (x)_k$. ■

Definition 5.11 (Bell numbers)

$B(n)$, the n-th **Bell number**, is the number of all partitions of $[n]$; $B(0) = 1$. So $B(n) = \sum_{i=0}^{n} S(n,i)$.

Theorem 5.12

For every nonnegative integer n:

$$B(n+1) = \sum_{i=0}^{n} \binom{n}{i} B(i). \tag{5.3}$$

Proof. In a partition of $[n+1]$, let i be the number of elements *not* in the block of $n+1$. These can be chosen in $\binom{n}{i}$ ways and partitioned in $B(i)$ ways; the rest of $[n]$ joins the block of $n+1$. Sum over i. ■

5.3 Integer partitions

Motivation. When neither balls nor boxes are distinguishable, only the multiset of box sizes remains: a partition of the integer n. There is no usable formula for $p(n)$, but Ferrers diagrams turn many identities into pictures, via conjugation, hook removal and adding a part 1.

Definition 5.13 (Integer partitions)

If $a_1 \geq a_2 \geq \cdots \geq a_k \geq 1$ are integers with $a_1 + \cdots + a_k = n$, then $(a_1, \ldots, a_k)$ is a **partition** of the integer n. $p(n)$ denotes the number of partitions of n, and $p_k(n)$ the number with exactly k parts.

("Set partition" and "integer partition" are different notions; the context, $[n]$ versus n, tells which is meant.)

Example 5.14

$p(5) = 7$: $(5), (4,1), (3,2), (3,1,1), (2,2,1), (2,1,1,1), (1,1,1,1,1)$.

No recurrence computes $p(n)$ simply from smaller values (but see the Notes). Asymptotically

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right), \tag{5.4}$$

faster than any polynomial and slower than any c^n, $c > 1$.

The **Ferrers shape** of $(x_1, \dots, x_k)$ has x_i unit boxes in row i, all rows left-justified; partitions of n and Ferrers shapes of size n correspond bijectively. Reflecting in the main diagonal gives the **conjugate** partition, whose i-th row is the i-th column of the original; the conjugate of $(4, 2, 1)$ is $(3, 2, 1, 1)$.

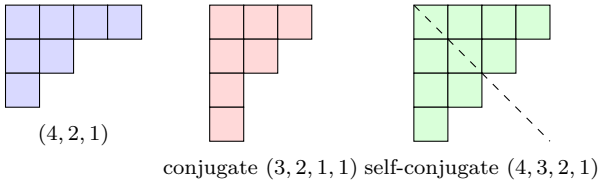


Figure. *Ferrers shapes. Conjugation reflects in the main diagonal (dashed); a self-conjugate shape is symmetric about it.*

Definition 5.15 (Self-conjugate)

A partition is **self-conjugate** if it equals its conjugate.

Example 5.16

$(4, 3, 2, 1)$, $(5, 1, 1, 1, 1)$ and $(4, 2, 1, 1)$ are self-conjugate.

Theorem 5.17

The number of partitions of n into at most k parts equals the number of partitions of n into parts not larger than k.

Proof. The first counts Ferrers shapes of size n with at most k rows, the second those with at most k columns; conjugation is a bijection between them. ∎

(For $n = 5$, $k = 2$: $(5), (4, 1), (3, 2)$ versus $(2, 2, 1), (2, 1, 1, 1), (1^5)$.)

Theorem 5.18

The number of partitions of n into distinct odd parts equals the number of self-conjugate partitions of n.

Proof.

Step 1 (the map). For self-conjugate $\pi = (\pi_1, \dots, \pi_t)$, remove the first row and first column of its Ferrers shape: by symmetry these are $2\pi_1 - 1$ boxes, the **hook** of the top left box. Repeat on what remains: the i-th hook has $2\pi_i - (2i - 1)$ boxes, as long as boxes remain. Let $f(\pi)$ be the sequence of hook lengths $(2\pi_1 - 1, 2\pi_2 - 3, \dots)$, a partition of n into odd parts. The parts are distinct, indeed strictly decreasing, since $\pi_1 \geq \pi_2 \geq \cdots$ gives $(2\pi_i - 2i + 1) - (2\pi_{i+1} - 2i - 1) = 2(\pi_i - \pi_{i+1}) + 2 \geq 2$.

Step 2 (inverse). Given distinct odd parts $\alpha = (2a_1 - 1, 2a_2 - 3, \dots, 2a_u - (2u - 1))$, any π with $f(\pi) = \alpha$ must have a_1 boxes in its first row and in its first column, a_2 further boxes in row and column 2, and so on. This determines a unique self-conjugate shape, and it has image α. So f is a bijection. ∎

Example 5.19

For $\pi = (6,6,4,3,2,2)$ (self-conjugate), $f(\pi) = (11,9,3)$: the three successive hooks have 11, 9 and 3 boxes.

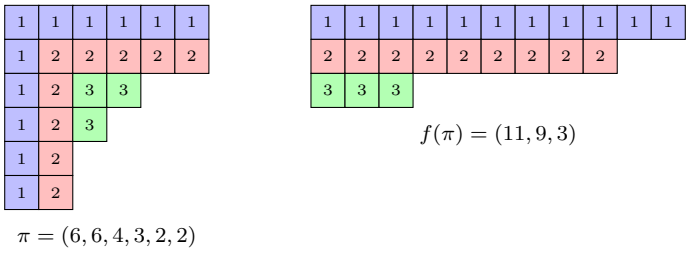


Figure. *Theorem 5.18: hook i of the self-conjugate shape (boxes labelled i) becomes row i of a partition into distinct odd parts.*

Theorem 5.20

Let $q(n)$ be the number of partitions of n with all parts at least 2. Then $q(n) = p(n) - p(n-1)$ for $n \geq 2$.

Proof. Appending a part 1 is a bijection from partitions of $n-1$ onto partitions of n having a part equal to 1. The partitions of n not obtained this way are exactly those counted by $q(n)$. ∎

Type of a set partition. Sorting the block sizes of a set partition π of $[n]$ decreasingly gives a partition of the integer n, the **type** of π.

Example 5.21

$\{1,5,6\},\{2,7\},\{3,9\},\{4,8\},\{10\}$ has type $(3,2,2,2,1)$.

Theorem 5.22

Let $a = (a_1, \ldots, a_k)$ be a partition of n and m_i the multiplicity of i as a part of a. The number of set partitions of $[n]$ of type a is

$$P_a = \frac{\binom{n}{a_1,\ldots,a_k}}{\prod_{i\geq 1} m_i!}.$$

Proof. Take a_j balls of colour j and order all n balls linearly, in $\binom{n}{a_1,\ldots,a_k}$ ways (Theorem 3.5). Put $i,j \in [n]$ in the same block iff positions i and j carry the same colour; this gives a set partition of type a, and every one arises. Two colourings give the same partition iff they differ by permuting colour classes of equal size, and the m_i classes of size i can be permuted in $m_i!$ ways. So each partition of type a arises exactly $\prod_i m_i!$ times. ∎

Example 5.23

There are $\binom{10}{3,2,2,2,1}/(1!\,3!\,1!) = 12600$ set partitions of $[10]$ of type $(3,2,2,2,1)$.

Example 5.24

The partitions of $[2n]$ into blocks of size 2 (perfect matchings) number

$$\frac{\binom{2n}{2,2,\dots,2}}{n!} = \frac{(2n)!}{2\cdot 4\cdots 2n} = \prod_{i=1}^{n}(2i-1) = (2n-1)!!.$$

	Balls	Boxes	Number
Surjections	n distinct	k distinct	$k!\,S(n,k)$
	n distinct	any number, distinct	$\sum_{i=1}^{n} i!\,S(n,i)$
Compositions	n identical	k distinct	$\binom{n-1}{k-1}$
	n identical	any number, distinct	2^{n-1}
Set partitions	n distinct	k identical	$S(n,k)$
	n distinct	any number, identical	$B(n)$
Integer partitions	n identical	k identical	$p_k(n)$
	n identical	any number, identical	$p(n)$
empty boxes allowed (k boxes fixed):			
Functions	n distinct	k distinct	k^n
Weak compositions	n identical	k distinct	$\binom{n+k-1}{k-1}$
Set partitions	n distinct	k identical	$\sum_{i=1}^{k} S(n,i)$
Integer partitions	n identical	k identical	$\sum_{i=1}^{k} p_i(n)$

Notes

With the **pentagonal numbers** $k(3k-1)/2$, $k\in\mathbb{Z}$ (that is $0,1,2,5,7,12,\dots$), MacMahon's recurrence reads

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + \cdots, \tag{5.5}$$

where the i-th term has sign $(-1)^{\lfloor i/2\rfloor}$; e.g. $p(8) = 15+11-3-1 = 22$. Euler's pentagonal number theorem: n has as many partitions into an even number of distinct parts as into an odd number, unless n is pentagonal, when the two counts differ by one.

5.4 Selected exercises

Motivation. The book exercises below compare Bell numbers with factorials, evaluate an alternating Stirling sum by a single substitution, and relate partitions with and without singleton blocks by a bijection.

Exercise 5.2 (Bell numbers versus factorials)

Prove that $B(n) < n!$ for $n \geq 3$.

Solution.

Step 1 (an injection). Given a partition π of $[n]$, define $\sigma_\pi : [n] \to [n]$ as follows: if x is not the largest element of its block, $\sigma_\pi(x)$ is the next larger element of the block; if it is the largest, $\sigma_\pi(x)$

is the smallest element of the block. On a block $\{y_1 < \cdots < y_r\}$ this is $y_1 \mapsto y_2 \mapsto \cdots \mapsto y_r \mapsto y_1$, so σ_π is a permutation of $[n]$ whose cycles are exactly the blocks of π. Hence π can be read off from σ_π, and $\pi \mapsto \sigma_\pi$ is injective. This gives $B(n) \le n!$.

Step 2 (not surjective). Let $n \ge 3$ and let τ be the permutation with $\tau(1) = 3$, $\tau(3) = 2$, $\tau(2) = 1$ and $\tau(x) = x$ for $x \ge 4$. If $\tau = \sigma_\pi$, the blocks of π would be the cycles of τ, so $\{1, 2, 3\}$ would be a block, and then $\sigma_\pi(1) = 2 \ne 3 = \tau(1)$. So τ is not in the image, and $B(n) \le n! - 1 < n!$. ■

Exercise 5.16 (An alternating sum of Stirling numbers)

Prove that for all positive integers n,

$$\sum_{k=1}^{n} (-1)^k k!\, S(n,k) = (-1)^n.$$

Explain the meaning of this identity in words.

Solution. Substitute $x = -1$ in Corollary 5.10, which holds for every real x. Since

$$(-1)_k = (-1)(-2)\cdots(-k) = (-1)^k k!$$

and $S(n,0) = 0$ for $n \ge 1$, we get $(-1)^n = \sum_{k=1}^{n} S(n,k)(-1)^k k!$, the claim.

Meaning: $k!\, S(n,k)$ is the number of *ordered* partitions of $[n]$ into k blocks (partitions whose blocks are put in a line), equivalently of surjections $[n] \to [k]$ (Corollary 5.9). So among the ordered partitions of $[n]$, those with an even number of blocks and those with an odd number of blocks differ by exactly one, and the even ones are in excess iff n is even. ■

Exercise 5.34 (Partitions without singleton blocks)

Let $F(n)$ be the number of partitions of $[n]$ with no singleton block. Prove that $B(n) = F(n) + F(n+1)$ for $n \ge 1$, by a bijection.

Solution. Split the partitions of $[n]$ into the set $\mathcal{P}_0$ of those without singleton blocks, $|\mathcal{P}_0| = F(n)$, and the set $\mathcal{P}_1$ of those with at least one singleton block. It suffices to give a bijection Φ from $\mathcal{P}_1$ onto the set $\mathcal{F}_{n+1}$ of partitions of $[n+1]$ without singleton blocks.

Step 1 (the map). For $\pi \in \mathcal{P}_1$ let $S \ne \emptyset$ be the union of its singleton blocks. Let $\Phi(\pi)$ be the partition of $[n+1]$ whose blocks are the non-singleton blocks of π together with $S \cup \{n+1\}$. The new block has at least two elements and the other blocks are unchanged, so $\Phi(\pi) \in \mathcal{F}_{n+1}$.

Step 2 (the inverse). For $\rho \in \mathcal{F}_{n+1}$, the block containing $n+1$ is $T \cup \{n+1\}$ with $T \ne \emptyset$, since it is not a singleton. Let $\Psi(\rho)$ be the partition of $[n]$ consisting of the other blocks of ρ and the singletons $\{t\}$, $t \in T$. Then $\Psi(\rho) \in \mathcal{P}_1$.

Step 3 (inverse maps). In $\Psi(\rho)$ the singleton blocks are exactly those coming from T, because the other blocks of ρ have at least two elements. Hence $\Phi(\Psi(\rho)) = \rho$. Conversely $\Psi(\Phi(\pi)) = \pi$ by construction. So Φ is a bijection, $|\mathcal{P}_1| = F(n+1)$, and $B(n) = |\mathcal{P}_0| + |\mathcal{P}_1| = F(n) + F(n+1)$. ■